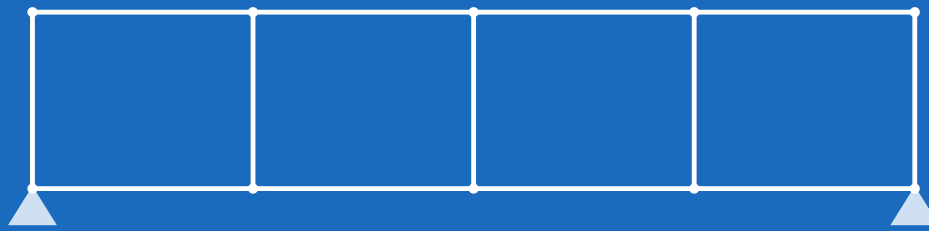
TRUSS REPORT

Vierendeel girder

Plane truss — analysis and CIRSOC 301 rod checks

Author A. Student · Course Structures I · 13/10/2026



MAX TENSION	MAX COMPRESSION	DEFLECTION	MOST USED ROD	BUCKLING Λ_1	WEIGHT
+40.8 kN	-40.8 kN	55.9 mm	0.55	144.35	2951 kg
rod 1	rod 5	L/358 · limit L/300	rod 9 · H1-1b, axial + bending	the load can grow ×144.3	carries 60.0 kN · 2× own weight

The model

Span	20.00 m
Height	4.00 m
Nodes / rods	10 / 13
Supports	pin at 0, rollerX at 4
Loads	3 nodal, total 60.0 kN
Rod families	1, with their sections:
Frame	SHS 200x200x8, F24 (CIRSOC) / A36

What the analysis found

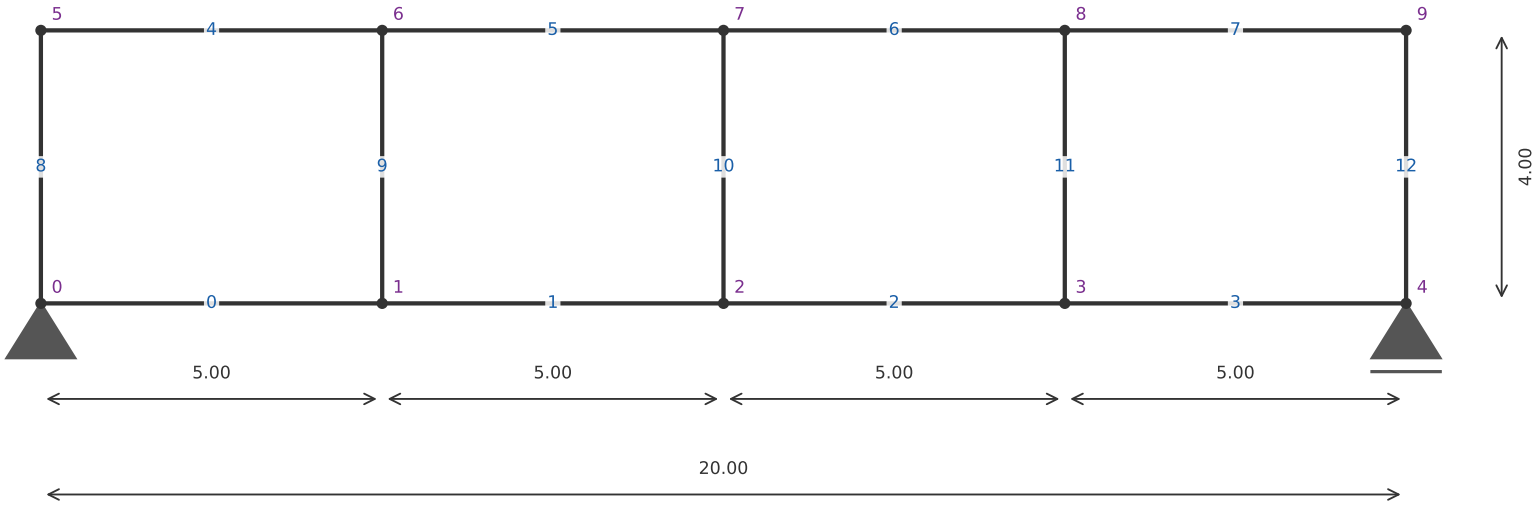
Stability	Statically indeterminate to degree 12 (deformations + r − DOF = 39 + 3 − 30 = 12): the forces also depend on the stiffness of the rods.
Most used rod	rod 9 at 0.55 (H1-1b, axial + bending)
Rods over capacity	0 of 13 checked
Buckling	$\lambda_1 = 144.35$
Shear and moment	present — see the Shear, Moment, Joints and Rod detail sheets
Joint moment	largest -43.44 kN·m at joint 3 (clockwise, from rod 11)

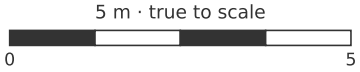
Contents

2 Geometry
3 Axial forces
4 Deflection and stress
5 Reactions
6 Utilisation
7 Buckling
8 Shear V
9 Moment M
10 Joints
11 Rods
12 Rod detail
13 Explain rod 9 and 11

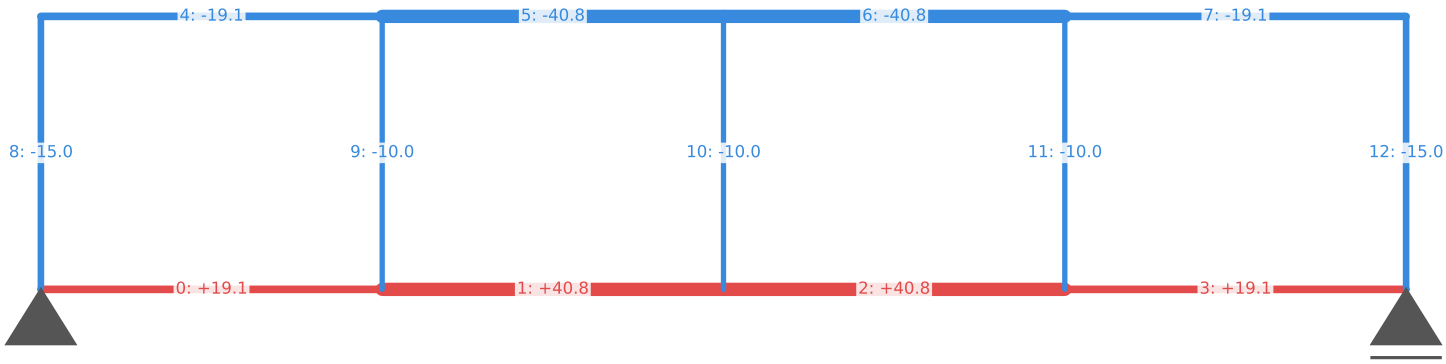
Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

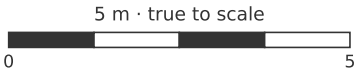
Node numbers in purple, rod numbers in blue. Dimensions in m.



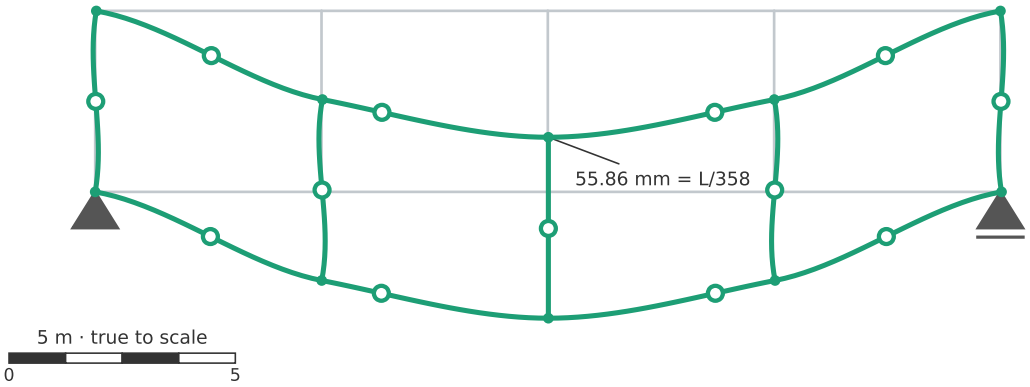


Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 40.8).



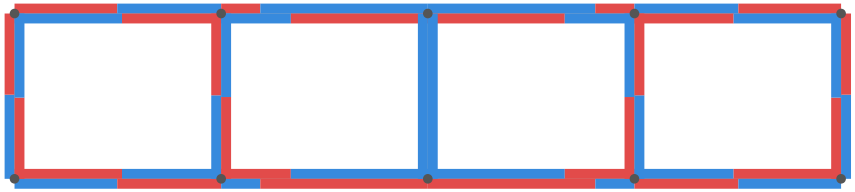


Elastic line: the deflected shape (green) over the truss (grey), every rigid rod drawn as the curve it bends into. Geometry to scale; displacements magnified × 50. ○ = point of contraflexure (M = 0), where a rod changes the way it bends: 13 of them. Largest vertical deflection 55.86 mm against the limit L/300 = 66.7 mm: OK.

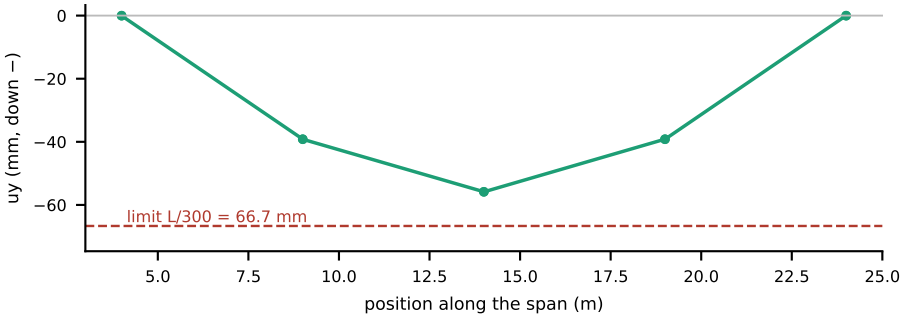


Fibre stress top and bottom face of each member

red = tension, blue = compression. Axial force and bending together (N/A ± M·c/I); a band flips colour at a point of contraflexure.

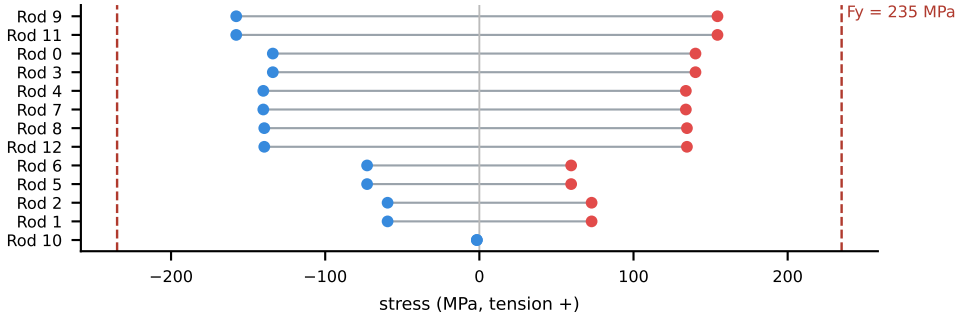


Deflection along the span vertical movement of every joint



Largest 55.86 mm at joint 7 = L/358. Within the limit.

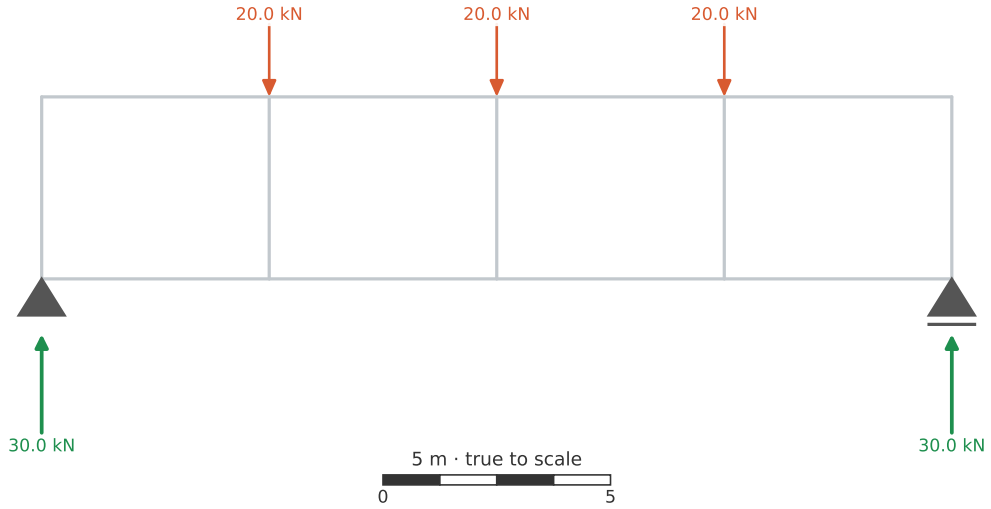
Fibre stress values the two faces of each rod, MPa



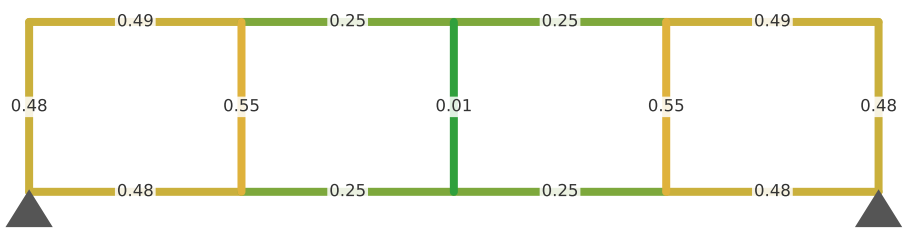
The 13 most stressed rods; the bar between a rod's two dots is its bending.

Support reactions (green arrows), as the supports push on the truss, against the loads (orange); Ry positive upwards. The sums balance the loads.

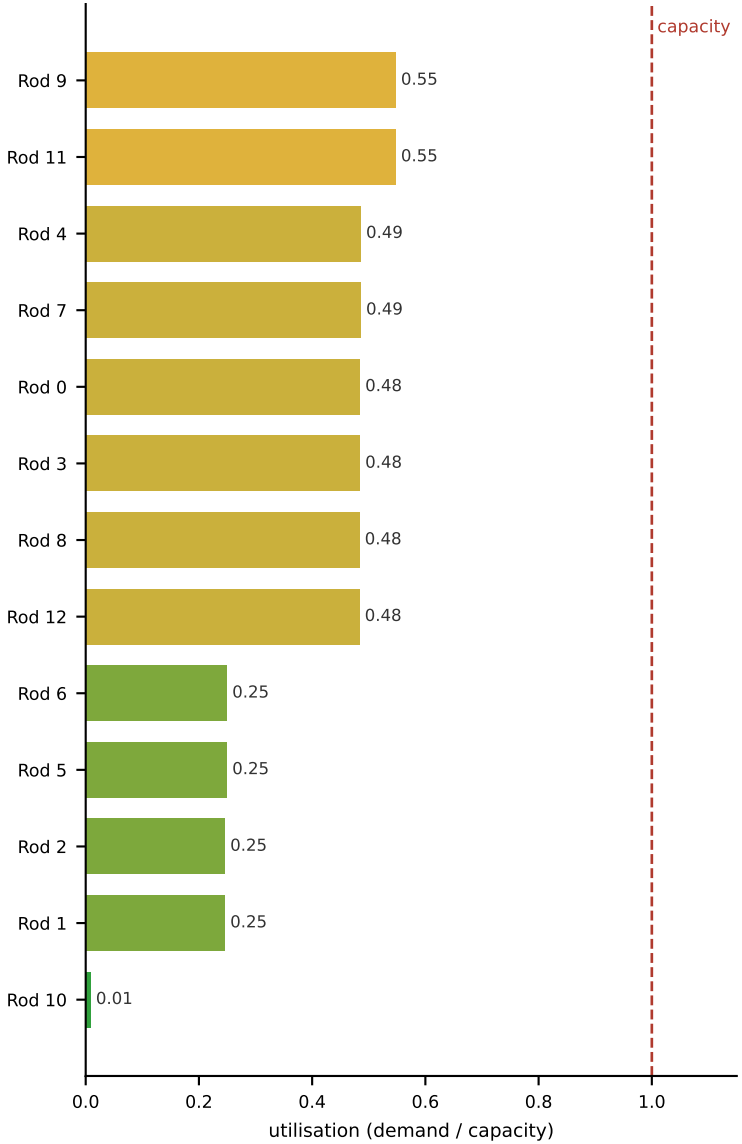
Node	Type	Rx (kN)	Ry (kN)
0	pin	0.00	30.00
4	rollerX	0.00	30.00
Σ		0.00	60.00

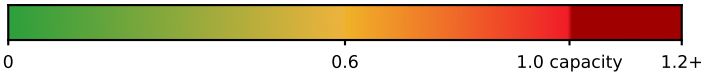


Utilisation: demand over CIRSOC 301 capacity. Most used rod 9 at 0.55 (H1-1b, axial + bending). Every checked rod is within capacity.



5 m · true to scale
0 5

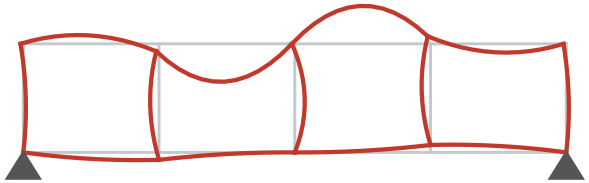




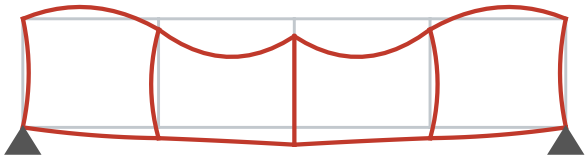
grey = no steel section, not checked

λ is the factor on the load as it stands. $\lambda_1 = 144.35$, so the load can grow 144.3× before it buckles. Mode 2 is 209.39, 45 % higher: one member decides the buckling load.

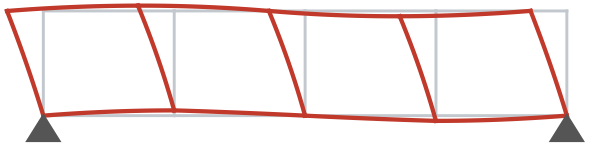
Mode 1: λ = 144.35
rod 5 bows

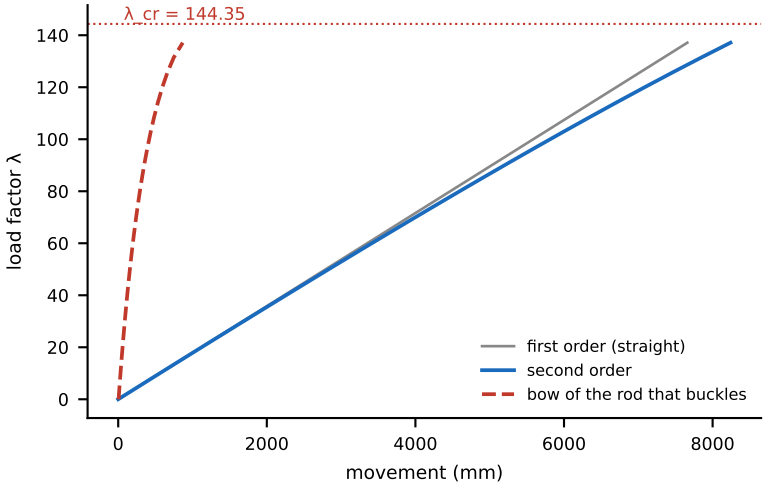


Mode 2: λ = 209.39
rod 5 bows



Mode 3: λ = 222.54
node 6 moves most

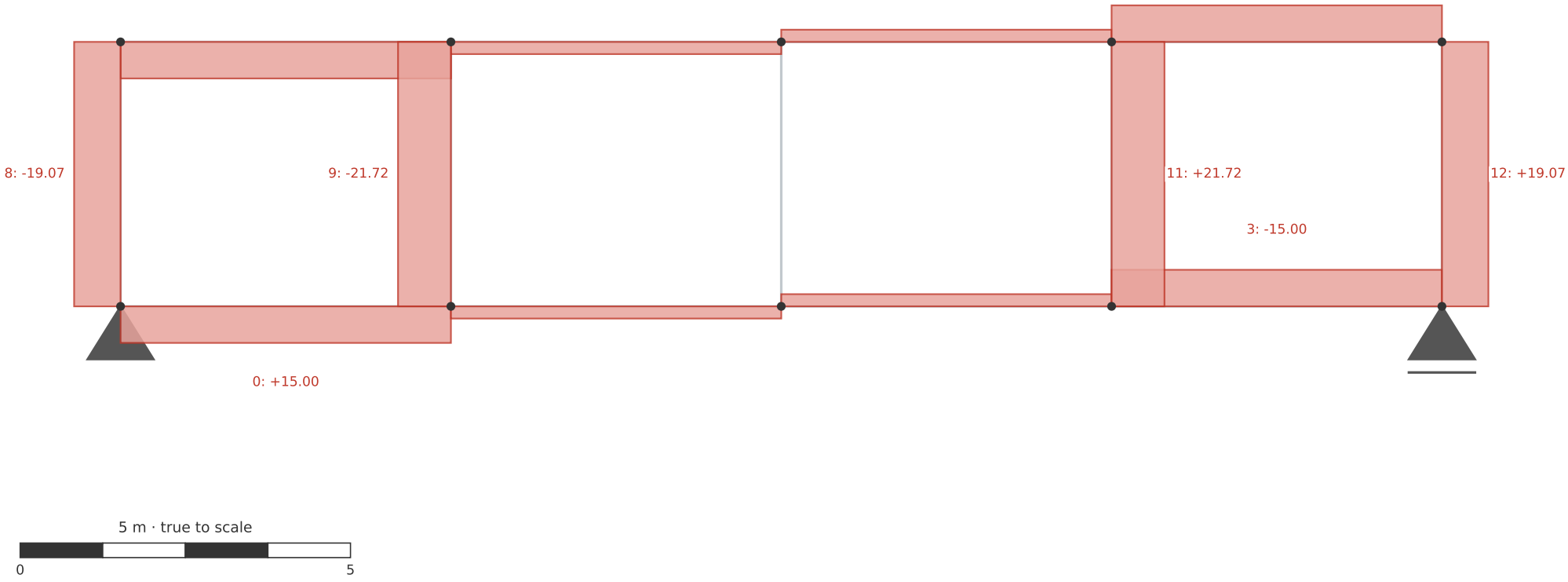




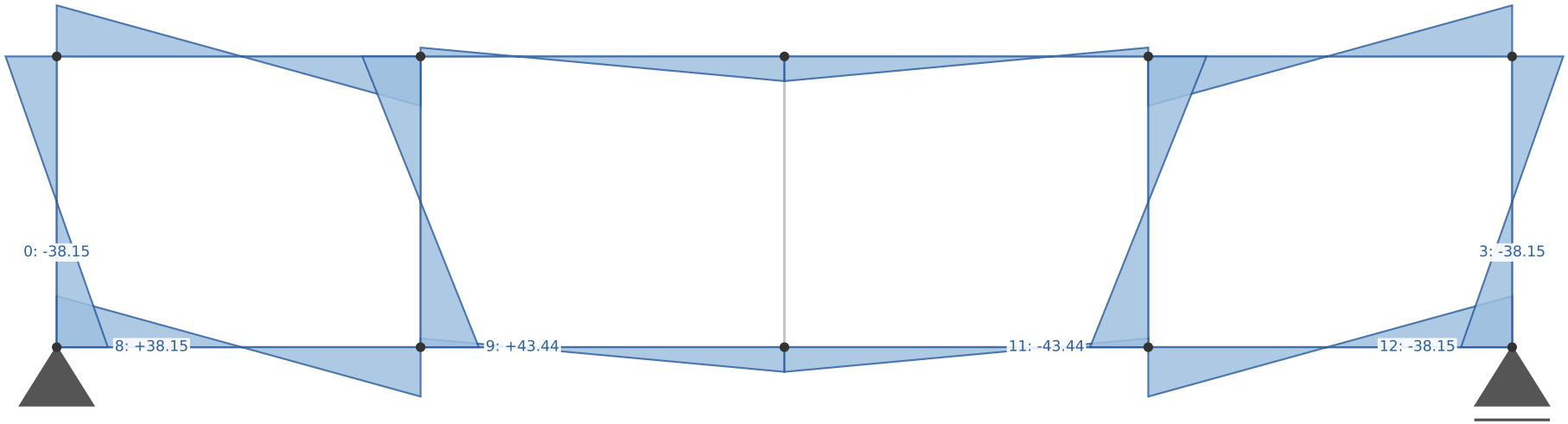
The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 5.00 mm in the first mode's shape.

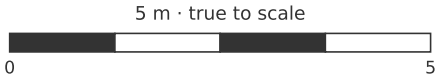
Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Shear force V along every rigid rod. Biggest +21.72 kN in rod 11; the band is drawn off each member, proportional to the value, and labelled on the six largest.

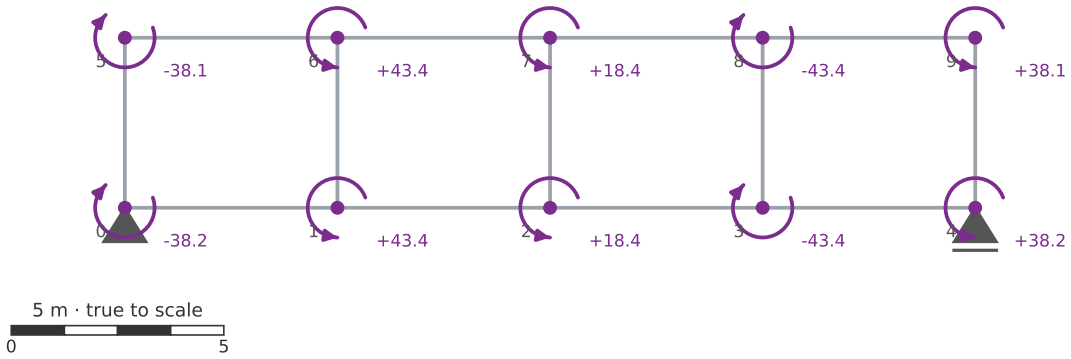


Bending moment M along every rigid rod, drawn on the face it puts in TENSION, as on screen (sagging positive: below a horizontal rod). Biggest -43.44 kN·m in rod 11; the band is drawn off each member, proportional to the value, and labelled on the six largest.





The moment at each joint: the largest one any single rod hands it -- what a rigid connection is designed for -- with its sign, which way it turns the joint (the arrow, anticlockwise = +, as on screen), and which face of that rod it puts in tension. Σ M is the check: ~0 at a free joint by equilibrium, the support's own moment at a fixed one.

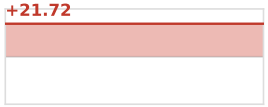
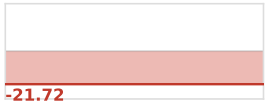
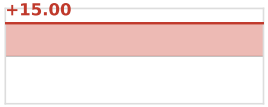
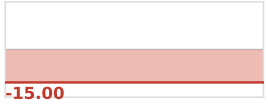
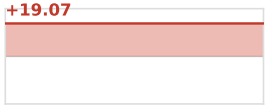
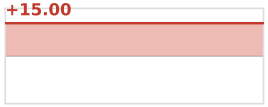
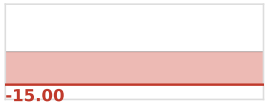


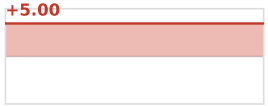
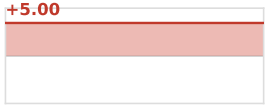
Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
3	rigid	43.44	−	clockwise	11	left	0.00 ✓
1	rigid	43.44	+	anticlockwise	9	right	0.00 ✓
8	rigid	43.43	−	clockwise	11	right	0.00 ✓
6	rigid	43.43	+	anticlockwise	9	left	0.00 ✓
0	rigid · pin	38.15	−	clockwise	0	top	0.00 ✓
4	rigid · rollerX	38.15	+	anticlockwise	3	top	0.00 ✓
5	rigid	38.14	−	clockwise	4	top	0.00 ✓
9	rigid	38.14	+	anticlockwise	7	top	0.00 ✓
7	rigid	18.42	+	anticlockwise	6	bottom	0.00 ✓
2	rigid	18.41	+	anticlockwise	2	bottom	0.00 ✓

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use
0	0–1	Frame	SHS 200x200x8	5.00	+19.07	H1-1b, axial + bending	0.48 OK
1	1–2	Frame	SHS 200x200x8	5.00	+40.79	H1-1b, axial + bending	0.25 OK
2	2–3	Frame	SHS 200x200x8	5.00	+40.79	H1-1b, axial + bending	0.25 OK
3	3–4	Frame	SHS 200x200x8	5.00	+19.07	H1-1b, axial + bending	0.48 OK
4	5–6	Frame	SHS 200x200x8	5.00	-19.07	H1-1b, axial + bending	0.49 OK
5	6–7	Frame	SHS 200x200x8	5.00	-40.79	H1-1b, axial + bending	0.25 OK
6	7–8	Frame	SHS 200x200x8	5.00	-40.79	H1-1b, axial + bending	0.25 OK
7	8–9	Frame	SHS 200x200x8	5.00	-19.07	H1-1b, axial + bending	0.49 OK
8	0–5	Frame	SHS 200x200x8	4.00	-15.00	H1-1b, axial + bending	0.48 OK
9	1–6	Frame	SHS 200x200x8	4.00	-10.00	H1-1b, axial + bending	0.55 OK
10	2–7	Frame	SHS 200x200x8	4.00	-10.00	inelastic buckling (E3-2)	0.01 OK
11	3–8	Frame	SHS 200x200x8	4.00	-10.00	H1-1b, axial + bending	0.55 OK
12	4–9	Frame	SHS 200x200x8	4.00	-15.00	H1-1b, axial + bending	0.48 OK

One row per rod that bends, worst first, down the left column and then the right. The thumbnail shows where the rod is; each diagram is scaled to its own rod, as on screen, with its largest value written at the peak. Rods with no shear or moment are in the Rods table.

	Where	Shear V (kN)	Moment M (kN·m)
Rod 11 use 0.55 3–8 · 4.00 m SHS 200x200x8 N -10.00 kN			
Rod 9 use 0.55 1–6 · 4.00 m SHS 200x200x8 N -10.00 kN			
Rod 0 use 0.48 0–1 · 5.00 m SHS 200x200x8 N +19.07 kN			
Rod 8 use 0.48 0–5 · 4.00 m SHS 200x200x8 N -15.00 kN			
Rod 3 use 0.48 3–4 · 5.00 m SHS 200x200x8 N +19.07 kN			
Rod 12 use 0.48 4–9 · 4.00 m SHS 200x200x8 N -15.00 kN			
Rod 4 use 0.49 5–6 · 5.00 m SHS 200x200x8 N -19.07 kN			
Rod 7 use 0.49 8–9 · 5.00 m SHS 200x200x8 N -19.07 kN			

	Where	Shear V (kN)	Moment M (kN·m)
Rod 6 use 0.25 7–8 · 5.00 m SHS 200x200x8 N -40.79 kN			
Rod 5 use 0.25 6–7 · 5.00 m SHS 200x200x8 N -40.79 kN			
Rod 2 use 0.25 2–3 · 5.00 m SHS 200x200x8 N +40.79 kN			
Rod 1 use 0.25 1–2 · 5.00 m SHS 200x200x8 N +40.79 kN			



Rod 9

utilisation 0.55

H1-1b, axial + bending

```
Rod 9
=====
Steel, CIRSOC 301-2018 -- A = 6,144 mm², Fy = 235 MPa, L = 4.000 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,000 mm / 78.5 mm
      = 51.0

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 51.0²
      = 759.30 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 51.0 vs 4.71 √(200,000/235) = 137.4
      = inelastic

 4. Critical stress  [E3-2]
      Fcr = 0.658^(Fy/Fe) · Fy
      = 0.658^(235/759.30) × 235
      = 206.45 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 206.45 × 6,144
      = 1,141.57 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 10.00 / 1,141.57
      = 0.009

 7. Bending capacity  [F.1]
      Mc = φb · Fy · S
      = 0.9 × 235 × 378,143 mm³
      = 79.977 kN·m

 8. Axial + bending  [H1-1b]
      Pr/(2Pc) + (Mry/Mc + Mrx/Mcw)
      = 0.009/2 + (0.000/79.977 + 0.000/79.977)
      = 0.548

 9. Governs
      the largest ratio
      = H1-1b, axial + bending
      = 0.548

utilisation 0.548 -- OK
```



Rod 11

utilisation 0.55

H1-1b, axial + bending

```
Rod 11
======
Steel, CIRSOC 301-2018 -- A = 6,144 mm², Fy = 235 MPa, L = 4.000 m

 1. Slenderness  [E2]
      K·L / r
      = 1 × 4,000 mm / 78.5 mm
      = 51.0

 2. Euler stress  [E3-4]
      Fe = π² E / (KL/r)²
      = π² × 200,000 / 51.0²
      = 759.30 MPa

 3. Which branch  [E3]
      KL/r ≤ 4.71 √(E/Fy) ?
      = 51.0 vs 4.71 √(200,000/235) = 137.4
      = inelastic

 4. Critical stress  [E3-2]
      Fcr = 0.658^(Fy/Fe) · Fy
      = 0.658^(235/759.30) × 235
      = 206.45 MPa

 5. Design strength  [E3-1]
      Pd = φc · Fcr · A
      = 0.9 × 206.45 × 6,144
      = 1,141.57 kN

 6. Compression ratio  [E3]
      |N| / Pd
      = 10.00 / 1,141.57
      = 0.009

 7. Bending capacity  [F.1]
      Mc = φb · Fy · S
      = 0.9 × 235 × 378,143 mm³
      = 79.977 kN·m

 8. Axial + bending  [H1-1b]
      Pr/(2Pc) + (Mry/Mc + Mrx/Mcw)
      = 0.009/2 + (0.000/79.977 + 0.000/79.977)
      = 0.548

 9. Governs
      the largest ratio
      = H1-1b, axial + bending
      = 0.548

utilisation 0.548 -- OK
```